

Overview of Semi-supervised learning

Subject: Semi-supervised learning

1.

$$\underset{f \in H}{\operatorname{argmin}} \left(\frac{1}{n} \sum_{i=1}^n V(f(x_i), y_i) + \lambda \|A\|_F^2 \|f\|_{H^2}^2 + \lambda \int_M |\nabla f(x)|^2 dp(x) \right)$$

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2.

Continuity / smoothness assumption Points that are close to each other are more likely to share a label. This is also generally assumed in supervised learning and yields a preference for geometrically simple decision boundaries. In the case of semi-supervised learning, the smoothness assumption additionally yields a preference for decision boundaries in low-density regions, so few points are close to each other but in different classes.

3.

History The heuristic approach of self-training (also known as self-learning or self-labeling) is historically the oldest approach to semi-supervised learning, with examples of applications starting in the 1960s. The transductive learning framework was formally introduced by Vladimir Vapnik in the 1970s. Interest in inductive learning using generative models also began in the 1970s. A probably approximately correct learning bound for semi-supervised learning of a Gaussian mixture was demonstrated by Ratsaby and Venkatesh in 1995.

4.

$$f^* = \underset{f \in H}{\operatorname{argmin}} \left(\sum_{i=1}^n (1 - y_i f(x_i))^2 + \lambda \|h\|_{H^2}^2 + \lambda \sum_{i=1}^n (1 - |f(x_i)|) + \lambda \sum_{i=1}^n (1 - y_i f(x_i))^2 + \lambda \|h\|_{H^2}^2 + \lambda \sum_{i=1}^n (1 - |f(x_i)|) \right)$$

5.

Laplacian regularization Laplacian regularization has been historically approached through graph-Laplacian. Graph-based methods for semi-supervised learning use a graph representation of the data, with a node for each labeled and unlabeled example. The graph may be constructed using domain knowledge or similarity of examples; two common methods are to connect each data point to its k nearest neighbors or to examples within some distance ϵ . The weight W_{ij} of an edge between x_i and x_j is then set to $e^{-\|x_i - x_j\|^2 / \epsilon^2}$. Within the framework of manifold regularization, the graph serves as a proxy for the manifold. A term is added to the standard Tikhonov regularization problem to enforce smoothness of the solution relative to the manifold (in the intrinsic space of the problem) as well as relative to the ambient input space. The minimization problem becomes

6.

$$\underset{\Theta}{\operatorname{argmax}} \left(\log p(\{x_i, y_i\}_{i=1}^l | \theta) + \lambda \log p(\{x_i\}_{i=1}^l | \theta) \right) + \lambda \log p(\{x_i\}_{i=1}^l | \theta)$$

7.

Low-density separation Another major class of methods attempts to place boundaries in regions with few data points (labeled or unlabeled). One of the most commonly used algorithms is the transductive support vector machine, or TSVM (which, despite its name, may be used for inductive learning as well). Whereas support vector machines for supervised learning seek a decision boundary with maximal margin over the labeled data, the goal of TSVM is a labeling of the unlabeled data such that the decision boundary has maximal margin over all of the data. In addition to the standard hinge loss $(1 - yf(x))_+$ for labeled data, a loss function $(1 - |f(x)|)_+$ is introduced over the unlabeled data by letting $y = \operatorname{sign}(f(x))$. TSVM then selects $f^*(x) = h^*(x) + b$ from a reproducing kernel Hilbert space H by minimizing the regularized empirical risk:

8.

In human cognition Human responses to formal semi-supervised learning problems have yielded varying conclusions about the degree of influence of the unlabeled data. More natural learning problems may also be viewed as instances of semi-supervised learning. Much of human concept learning involves a small amount of direct instruction (e.g. parental labeling of objects during childhood) combined with large amounts of unlabeled experience (e.g. observation of objects without naming or counting them, or at least without feedback). Human infants are sensitive to the structure of unlabeled natural categories such as images of dogs and cats or male and female faces. Infants and children take into account not only unlabeled examples, but the sampling process from which labeled examples arise.

9.

An exact solution is intractable due to the non-convex term $(1 - |f(x)|)_+$, so research focuses on useful approximations. Other approaches that implement low-density separation include Gaussian process models, information regularization, and entropy minimization (of which TSVM is a special case).

10.

Weak Supervision in Predictive Maintenance Weak supervision is an emerging machine learning approach in predictive maintenance that addresses the challenge of limited or imprecise labeled data. Traditional predictive maintenance models often rely on large volumes of accurately labeled failure and operational data, which can be costly or impractical to obtain. Weak supervision mitigates this by leveraging imperfect sources of supervision—such as noisy labels, heuristics, domain expert rules, or partially labeled datasets—to train predictive models. By incorporating techniques such as data programming, label modeling, and semi-supervised learning, weak supervision enables the development of robust predictive maintenance systems capable of identifying equipment failures or

anomalies with reduced reliance on high-quality labeled data. This approach is particularly valuable in industrial settings where machine failures are rare and labeled fault data is scarce .

11.

Assumptions In order to make any use of unlabeled data, some relationship to the underlying distribution of data must exist. Semi-supervised learning algorithms make use of at least one of the following assumptions: